

CLASSCONNECT

Technical Information and Engineering Handover

A verified guide to the application architecture, frontend, backend, database, security, AI, documents, analytics, performance, testing, and deployment.

PRODUCT	LIVE APPLICATION
ClassConnect	class-connect-six.vercel.app
Punch line	Connecting the classroom beyond the classroom.
Repository	github.com/yugmakwana1811/ClassConnect
Snapshot	Commit ca27174 - Make workspace navigation feel instant

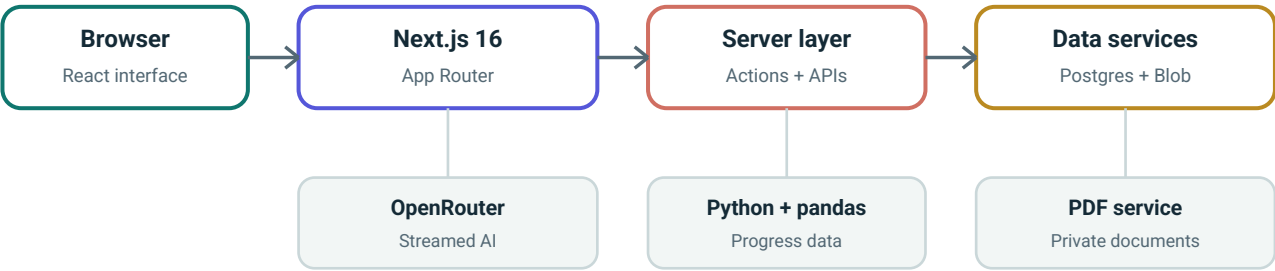
Prepared from direct repository, schema, route, test, Git, and deployment inspection

14 August 2026 | Asia/Kolkata

SECTION 01

Executive summary

ClassConnect is a production-deployed classroom workspace for teachers, students, and parents. It combines classroom operations, secure academic files, assessment review, learning analytics, parent visibility, and streamed AI assistance in one full-stack application.



Architecture in one sentence
The browser uses a React interface served by Next.js; authenticated server code reads and writes PostgreSQL through Prisma, protects private files in Vercel Blob, streams OpenRouter output, and loads pandas analytics independently.

Production identity

Item	Verified value
Product	ClassConnect
Punch line	Connecting the classroom beyond the classroom.
Production	https://class-connect-six.vercel.app/
Git repository	https://github.com/yugmakwana1811/ClassConnect
Current branch	agent/ship-edugrade-feature-updates
Current commit	ca271746e72b6df96ce4d43dfb42664030d3525d

Scope note
This document describes the current repository state. The internal npm package name remains edugrade-ai for continuity, but all visible branding is ClassConnect.

SECTION 02

Technology stack and repository

The application is a Next.js monolith: frontend and backend are deployed together, with a focused Python function for pandas aggregation.

Area	Technology	Purpose
Frontend	React 19.2.4 + Next.js 16.2.11	Server-rendered pages and interactive client components
Backend	Next.js Server Actions + Route Handlers	Authenticated reads, mutations, APIs, uploads, files, PDFs, and AI
Runtime	Node.js 24	Vercel Fluid Compute and local execution
Language	TypeScript	Strictly typed application code
Database	Neon PostgreSQL	Persistent users, classes, work, results, analytics, and activity
ORM	Prisma 6.19	Schema, migrations, relations, and queries
Storage	Private Vercel Blob	Question files, resources, and answer pages
AI	OpenRouter	Server-only free-model routing and streamed generation
Analytics	Python + pandas 3.0 + psycopg	Teacher progress aggregation
PDF	pdf-lib + fontkit + Roboto	Authorized A4 academic documents
Validation	Zod 4	Input and workflow validation
Testing	Vitest + ESLint + TypeScript	Regression, static, and production checks

Repository layout

Directory	Responsibility
src/app	Pages, layouts, APIs, Server Actions, loading states, and error boundaries
src/components	Shared UI, forms, navigation, PDF controls, charts, and test lockdown
src/lib	Authentication, AI, analytics, storage, PDFs, validation, and utilities
prisma	Database schema, migrations, and deterministic demo seed
api	Python student-progress Vercel Function
scripts	Demo verification and QA helpers
docs	Feature, testing, and deployment guidance
.github/workflows	CI quality gate

The inspected application surface contains approximately 204 relevant source and configuration files: 150 TypeScript/TSX files, three Python files, three SQL migration files, plus CSS, JavaScript, documentation, and assets.

SECTION 03

Frontend and user experience

React Server Components provide data-heavy screens, while Client Components are limited to browser interaction. This keeps the shipped JavaScript focused without losing responsiveness.

Shared workspace

- Persistent role-specific sidebar, top bar, mobile navigation, account access, and logout.
- Responsive custom design system with cards, tables, empty states, progress indicators, dialogs, and section-level loading states.
- Accessible show-password controls on sign-in and registration.
- Originkit Typewriter and Text Morph, plus React Bits ShinyText, integrated into the landing and AI presentation areas.
- The user-facing motion on/off preference was removed; controlled component animation remains where it is part of the visual design.

Role modules

Teacher	Student	Parent
Dashboard and class management	Dashboard and joined classes	Linked-student overview
Assignments, tests, worksheets	Assigned work and downloads	Marks and published results
Quiz builder and attempts	Quizzes and test mode	Quiz performance
Review Work, marks, feedback	Answer-page submission	Attendance
Resources and announcements	Published feedback and results	Announcements
Attendance and analytics	Personal analytics	AI Studio history
Student profiles and AI tools	AI doubt and revision help	Compact activity timeline

Class management

Every class card can open the class, start a new assignment, rename the class, or permanently delete it. Deletion requires exact-name confirmation. Database relations cascade correctly, and uploaded file cleanup is attempted for affected records.

Responsive behavior
Teacher, student, and parent workspaces are designed for desktop, laptop, tablet, and mobile widths. On mobile, core role navigation remains immediately accessible.

SECTION 04

Instant navigation and performance

The latest optimization focused on perceived speed first: immediate click feedback, immediate destination shells, cached revisits, background synchronization, and isolated secondary data.

20 ms Assignments response after optimization	14 ms Classes response after optimization	10 ms Warm assignment revisit	-66% Assignment query payload
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What changed

- Pointer, touch, and keyboard activation immediately updates navigation feedback.
- Likely routes are prefetched using Next.js navigation and hover, focus, or touch intent.
- Visited modules reuse the Next.js client cache and silently revalidate when appropriate.
- Dynamic route cache lifetime is 90 seconds; static route cache lifetime is 300 seconds.
- Route-aware loading shells replace blank pages and full-screen blockers.
- Slow analytics, graphs, and secondary information load independently.
- Independent database operations run in parallel where consistency permits.
- Assignment queries select only fields needed by the current screen.
- Submission completion uses client navigation instead of a full browser reload.
- Mutations refresh affected paths; logout clears role-specific cached content.

Measured behavior under delay

With an artificial one-second data delay, click feedback appeared in about 7 ms, the destination URL and page shell appeared in about 11 ms, and fresh data arrived later without freezing the interface.

Cache safety
Private classroom APIs are not persisted in a service worker or localStorage. Caching remains bound to the authenticated browser session and Next.js route cache, reducing cross-user leakage risk.

SECTION 05

Backend and API design

Next.js handles the application backend directly. There is no separate Express, NestJS, Django, Laravel, Java, GraphQL, Redis, or background-queue service.

Backend patterns

- Server Components perform authenticated server-side reads.
- Server Actions perform validated writes and role-protected workflow transitions.
- Route Handlers deliver AI streams, uploads, private files, PDFs, answer pages, and health checks.
- Prisma provides typed PostgreSQL access and transaction support.
- Python Fluid Compute serves only the pandas progress aggregation endpoint.

HTTP endpoints

Route	Purpose	Access
/api/ai/generate	Stream teaching or student-help output	Teacher or permitted student types
/api/ai/feedback	Stream teacher feedback suggestions	Teacher
/api/files/[kind]/[id]	Deliver private assignment/resource files	Authorized class member/teacher
/api/pdfs/[kind]/[id]	Preview or download generated documents	Role and relationship scoped
/api/submission-pages/[id]	Deliver protected answer pages	Owning student or class teacher
/api/uploads	Authorize controlled direct uploads	Enrolled student with open work
/api/student-progress	Return pandas progress aggregation	Teacher
/api/health	Check production dependencies	Operational endpoint

Error handling
Protected APIs distinguish unauthenticated, unauthorized, not-found, invalid-input, and unavailable-service conditions. Production responses avoid exposing stack traces and provider details.

SECTION 06

Database model and data integrity

PostgreSQL stores the complete classroom state. Prisma defines 23 related models and versioned migrations.

Domain	Models
Identity	User, AuthThrottle, Session, TeacherProfile, StudentProfile, ParentProfile
Relationships	ParentStudent, ClassRoom, ClassEnrollment
Coursework	Assignment, AssignmentAttachment, Submission, SubmissionPage
Assessment	Result, Feedback, Quiz, QuizQuestion, QuizAttempt
Classroom	Resource, Announcement, AttendanceRecord
Intelligence	AIContentGeneration, ActivityLog

Integrity rules

- A student cannot be enrolled in the same class twice.
- A student has at most one submission per assignment.
- Attendance is unique per class, student, and date.
- Parent-student relationships cannot be duplicated.
- Marks are validated against maximum marks.
- Class deletion cascades through dependent database records.
- Indexes cover frequent teacher, class, student, status, and date filters.

Workflow states

The schema supports teacher/student roles, draft/published/closed assignments, draft/submitted/reviewed/published submissions, present/absent/late/excused attendance, and typed AI generation records.

Migrations

Migration	Purpose
20260722000000_baseline	Versioned production baseline
20260806220000_add_parent_module	Parent profiles and student linking
20260806231500_add_query_indexes	High-frequency query indexes

SECTION 07

Authentication, authorization, and security

Authentication is custom and database-backed. Security checks are applied in layouts, Server Actions, API routes, file delivery, and database queries.

Credentials and sessions

- bcrypt password hashing with cost factor 12.
- Passwords require at least 10 characters, uppercase, lowercase, a number, and a special character; maximum length is 128.
- Sessions use 32-byte random tokens; only HMAC-SHA256 token hashes are stored.
- Sessions expire after 14 days and use HTTP-only, Secure-in-production, SameSite Lax cookies.
- Password or verified email changes invalidate other sessions.

Abuse protection

Five failed attempts within a 15-minute window trigger a 15-minute block. The throttle key incorporates operation scope, normalized identity, client IP, and a server secret.

Authorization boundaries

Protected information	Enforcement
Teacher solutions and marking schemes	Teacher role plus class ownership
Student submissions	Owning student or class teacher
Parent progress	Active parent-student relationship
Private resources	Classroom enrollment or teacher ownership
AI tools	Role plus allowed content type
PDF versions	Server-side teacher/student/parent policy

Production headers

Content Security Policy, HSTS, X-Content-Type-Options, frame controls, referrer policy, permissions policy, and Cross-Origin-Opener-Policy are configured globally. PDF routes permit same-origin preview while remaining sandboxed.

Secret handling
Database, authentication, Blob, Server Action, and OpenRouter credentials stay server-only. No secret values are committed or exposed to the browser.

SECTION 08

Test lockdown and AI assistance

Two specialist systems support assessments: a test-only fullscreen control layer and a server-streamed, role-restricted AI service.

Test-only lockdown

- Requires fullscreen and hides or disables workspace navigation.
- Blocks context menu, copy, cut, paste, drag, common new-tab shortcuts, and in-page window.open calls.
- Pauses when the tab is hidden, the window loses focus, or fullscreen is exited.
- Records interruption counts and warns before leaving.
- Removes all restrictions after submission and does not activate outside designated tests.

Browser limitation

A website cannot control operating-system switching, browser chrome, or a second physical device. The implementation provides strong browser-level and in-app enforcement without claiming impossible control.

AI configuration

Item	Current behavior
Primary free model	google/gemma-4-31b-it:free
Fallback free model	nvidia/nemotron-3-super-120b-a12b:free
Transport	Node.js ReadableStream using newline-delimited JSON
Visibility	Provider and model metadata remain server-side
Timeout	240 seconds
Output limit	4,096 tokens
Failure mode	Provider failover before output; deterministic safe templates when unavailable

AI permissions and safety

Students are limited to doubt help, revision help, and explanations. Parents cannot call AI generation. Automated feedback prompts exclude student names and answer content. Teachers remain responsible for review, marks, and publication; AI does not autonomously grade handwriting.

SECTION 09

Private storage and PDF documents

Academic files remain private, while a reusable PDF service generates polished student and teacher documents on authorized requests.

Private file storage

- Production uses private Vercel Blob through OIDC or a private read-write token.
- Local development can use a protected filesystem store when Blob is absent.
- Random UUID paths, filename sanitization, resolved-root checks, and content-type validation prevent predictable or traversal-based access.
- Student uploads require enrollment, a published assignment, an open submission, and a permitted storage prefix.
- Server uploads are capped at 3 MB; direct answer pages at 10 MB each; a submission supports up to 20 ordered pages.

PDF capabilities

Supported	Design and protection
Assignments, tests, quizzes, worksheets	A4, title hierarchy, class/subject, dates, duration, marks, instructions
Homework, revision, practice work	Reusable sections, spacing, page breaks, and print-safe colors
Submissions, reports, result summaries	Role and relationship scoped server generation
Teacher feedback and answer keys	Teacher-only solutions, explanations, rubrics, and marking guidance
Student versions	Questions and instructions without answers

Demo filenames

A deterministic helper maps seeded academic documents to Demo.pdf, Demo 1.pdf, Demo 2.pdf, and onward. Academic titles remain descriptive inside ClassConnect. There are 116 unique demo document mappings, generated on demand rather than exposed as public files.

Preview model
PDFs use an authorized same-origin route and native browser preview. Downloads preserve safe filenames; teacher-only material cannot be reached through student routes.

SECTION 10

Class 12 Commerce demonstration

The deterministic, idempotent seed creates a realistic classroom that can be reseeded without duplicating records.

Students and performance profiles

Student	Demonstration profile
Arjun Mehta	Existing Class 12 Commerce learner
Aarav Patel	High performer
Diya Shah	Above average
Riya Desai	Consistent average
Kabir Mehta	Needs improvement
Vivaan Joshi	Improving over time

Subjects

Accountancy, Economics, Business Studies, Informatics Practices, and English.

Seeded academic volume

Content	Count	Content	Count
Assignments	5	Tests	3
Quizzes	5	Worksheets	3
Practice activities	2	Homework activities	2
Revision resources	2	Project	1
Classwork activity	1	Supporting resources	Additional

Realistic variation

Seeded records include different completion levels, submission timing, late work, in-progress work, published and unpublished results, quiz attempts, attendance, concise teacher feedback, AI activity, announcements, and activity logs. Dashboard and analytics figures are calculated from these records instead of being hardcoded separately.

Demo integrity

Seed operations use deterministic identifiers and upserts. Re-running the seed does not create duplicate students, class enrollments, assessments, or PDF mappings.

SECTION 11

Analytics and progress graph

Teacher, student, and parent insights derive from persistent records. Secondary analytics are isolated so they cannot block the primary page experience.

Teacher analytics

- Class average from published assignment results.
- Completion rate from expected enrollment-assignment combinations.
- Pending and reviewed submission counts.
- Highest and lowest published scores.
- Attendance rate from attendance records.
- Topic averages with evidence counts.
- Quiz accuracy and students needing attention.

Pandas student-progress graph

The teacher-only Python endpoint authenticates the session, queries students from classes owned by that teacher, and combines published assignment results with completed quiz attempts. pandas orders records, clips scores to 0-100, removes duplicate assessment entries, selects up to 12 recent activities, and calculates averages and trends.

Layer	Responsibility
PostgreSQL	Teacher-scoped roster and assessed records
pandas	Cleaning, ordering, deduplication, averaging, trend classification
React SVG	Responsive lines, points, labels, legends, and accessible description
Progress cards	Average, assessed count, and Improving/Steady/Needs support status

Students without assessed work remain visible rather than disappearing from the chart. The graph fetches independently with a local loading and error state, so a Python cold start or temporary failure does not freeze the analytics screen.

SECTION 12

Quality assurance and deployment

The current application passed the complete local quality gate and was deployed successfully to Vercel.

Verified checks

Check	Result
Prisma schema validation	Passed
Strict TypeScript type checking	Passed
ESLint	Passed
Vitest	17 files, 89 tests passed
Production Next.js build	Passed
Vercel production deployment	Ready

Automated coverage

Tests cover AI routing and fallback, streaming, analytics, class codes, demo data, student insights, parent access, PDF generation, PDF filenames, document authorization, private storage, runtime configuration, submission access, test lockdown, validation, and workflow rules.

Production services

Service	Production role
Vercel Fluid Compute	Next.js Node.js 24 and Python functions
Neon PostgreSQL	Primary relational database
Private Vercel Blob	Protected academic files and answer pages
OpenRouter	Server-streamed free-model AI
GitHub Actions	Pull-request and main-branch quality gate

Live health

The production health endpoint reported ready with database connected and authentication, storage, Server Actions, and AI configured.

SECTION 13

Operations, environment, and limitations

The final section captures the exact operating requirements and the boundaries that should remain explicit.

Environment variables

Variable	Purpose
DATABASE_URL	TLS-enabled pooled PostgreSQL connection
AUTH_SECRET	At least 32 random bytes for session and throttle HMAC
NEXT_SERVER_ACTIONS_ENCRYPTION_KEY	Stable base64 key shared by production instances
OPENROUTER_API_KEY	Optional server-only live AI credential
BLOB_READ_WRITE_TOKEN	Private Blob token outside integrated Vercel OIDC
BLOB_STORE_ID	Automatically supplied by the Vercel Blob integration

Verified commands

```
npm ci
npm run db:deploy
npm run db:seed
npm run db:verify-demo
npm run dev
npm run typecheck
npm run lint
npm test
npm run build
npm start
npx vercel --prod --yes
```

Current limitations

- No automated email password recovery because no transactional email service is configured.
- Notifications are in-app; email, SMS, and push delivery are not configured.
- Quiz duration is a pacing recommendation rather than a server-enforced countdown.
- Browser lockdown cannot control the operating system or a second physical device.
- Free OpenRouter routes can experience provider-side capacity or latency limits.
- The application is a classroom workspace, not a full school SIS, billing, transport, payroll, or fee platform.
- The internal package name and legacy session cookie retain EduGrade identifiers, but visible branding is ClassConnect.

Vercel tooling

The installed Vercel CLI is behind version 59.0.0. Upgrade with `npm i -g vercel@latest` for current compatibility and tooling improvements.

Key repository references

[README.md](#) | [docs/FEATURES.md](#) | [docs/DEPLOYMENT.md](#) | [docs/TESTING.md](#) | [prisma/schema.prisma](#) | [prisma/seed.ts](#) | [src/lib/pdf-document-service.ts](#) | [src/lib/ai-routing.ts](#) | [api/student-progress.py](#)

CLASSCONNECT

Connecting the classroom beyond the classroom.

A secure, role-aware classroom workspace designed around real teaching, learning, assessment, and family visibility.

Verified technical handover

This document reflects the inspected repository snapshot, its production architecture, implemented workflows, quality checks, and stated limitations.

Frontend / Backend / PostgreSQL / Security / AI / PDF / Analytics / Vercel